

# S.E.D.A.R.

## Seismic Eddy Current and Active Damping Response

**Objective:** Demonstrate an active seismic damping system that detects building motion and reduces oscillation using controlled electromagnetic damping.

### 1. List of Components

| No. | Component                                         | Qty.      | Purpose                                                                   |
|-----|---------------------------------------------------|-----------|---------------------------------------------------------------------------|
| 1   | Arduino Nano + USB cable                          | 1         | Main controller. Reads the motion sensor and controls the damping system. |
| 2   | MPU6050 motion sensor                             | 1         | Measures acceleration and rotational movement of the model.               |
| 3   | IRF520 MOSFET switch module                       | 1         | Lets the Arduino switch the higher-current 12 V load.                     |
| 4   | 12 V DC electromagnets (P25/20 type)              | 2         | Create the controllable magnetic field for the damping demonstration.     |
| 5   | 12 V, 3 A DC adapter                              | 1         | Supplies power to the electromagnetic load.                               |
| 6   | Model building / frame                            | 1         | Mechanical structure used to demonstrate building oscillation.            |
| 7   | Conductive moving plate / metal element           | 1         | Moves relative to the magnetic field so eddy currents can oppose motion.  |
| 8   | Wires, connectors and breadboard / terminal block | As needed | Electrical connections and prototyping.                                   |
| 9   | Laptop / PC + USB connection                      | 1         | Optional. Displays sensor readings and motion data.                       |

### 2. Working Principle

SEDAR uses a **sensor + controller + electromagnetic damping system**. When the model moves, the MPU6050 detects the motion. The Arduino Nano processes the sensor signal and controls the electromagnets through the MOSFET.

### 3. Step-by-Step Working

- 1. Seismic motion** — A controlled vibration is applied to the model to represent ground motion.
- 2. Motion detection** — The MPU6050 measures the model's acceleration and rotational movement.
- 3. Signal processing** — The Arduino Nano receives the sensor data and detects significant motion.
- 4. Damping command** — When the motion reaches the programmed condition, the Arduino sends a control signal to the MOSFET.
- 5. Electromagnetic action** — The MOSFET switches the 12 V electromagnetic load.
- 6. Eddy-current damping** — Relative motion between the conductive plate and magnetic field induces eddy currents. Their magnetic effect opposes the motion.
- 7. Reduced oscillation** — Mechanical energy is dissipated mainly as heat in the conductor, reducing the vibration amplitude.
- 8. Feedback** — The sensor keeps measuring the motion so the controller can respond continuously.

### 4. Simple System Flow

Ground vibration → MPU6050 → Arduino Nano → IRF520 MOSFET → Electromagnetic damping system  
→ Eddy-current opposing force → Reduced oscillation

### 5. Important Engineering Note

For true eddy-current damping, there must be **relative motion between a conductive material and a magnetic field**. A stationary holding electromagnet alone does not automatically create eddy-current damping. The model must therefore be arranged so that the conductive moving plate changes position relative to the magnetic field.

**Safety:** Use insulated wiring and a correctly rated power supply. The 12 V electromagnetic load can draw significant current and may become warm.